

Orchestration Report

Status: 4 completed, 1 incomplete/cancelled

Session: 6a419a72d163a13dc7ec9177

Generated: 2026-06-29T01:35:48.695Z

Overview

Metric	Value
Total Tasks	5
Completed	4
Failed	0
Pending	0
Tool Calls	147
Errors	2
Duration	13m 49s

Agent Breakdown

Agent	Tasks
orchestrator	47
researcher	26
tuner	25
deployer	49

Summary

Status: 4 completed, 1 incomplete/cancelled

Duration: 13m 49s

Tasks: 4 completed, 0 failed, 1 cancelled, 0 pending

Tool calls: 147 (2 tool errors)

- Errors by tool: mongodb_find(1), get_details(1)

Findings

Results (4)

[orchestrator] Execute skill: EMULATOR_SPEC

Now I have the per-archetype score breakdown. This is very revealing:

Archetype	Adversarial	Avg Score	Max Score	Min Score
Honest	No	0.981	1.000	0.967
HonestTrader	No	0.973	0.987	0.953
Copycat	Yes	0.745	0.756	0.733
CopyTrader	Yes	0.725	0.732	0.718
Overfitter	Yes	0.600	0.605	0.593
SingleMetricGamer	Yes	0.593	0.601	0.577
Random	Yes	0.537	0.550	0.517
Sybil	Yes	0.393	0.406	0.377

The **Copycat** (0.745) and **CopyTrader** (0.725) are the highest-scoring adversaries. The separation gap is:

- Honest mean: $(0.981 + 0.973) / 2 = 0.977$
- Best adversarial: **0.745** (Copycat)
- Actual separation: $0.977 - 0.745 = 0.232$ (matches the ~0.244 empirical)

The surrogate model reported separation = 0.963, which would require best_adversary ≈ 0.018. The actual best adversary scores **0.745** — a **40x discrepancy** in the adversary score estimate.

Let me check the tuner optimization data to understand how the surrogate computed separation:

Used 46 tool call(s): list_memory_keys, mongodb_list_collections, read_memory, read_memory, read_memory, mongodb_find_one, mongodb_find_one, mongodb_find_one, list_tasks, list_tasks, mongodb_find_one, mongodb_find, mongodb_find, mongodb_find, mongodb_find, mongodb_find, write_memory, write_task, write_task, mongodb_find_one, mongodb_find, get_details, list_tasks, read_task, read_task, mongodb_find, mongodb_find, mongodb_find, cancel_task, get_details, mongodb_find_one, mongodb_find_one, write_task, write_task, write_memory, list_tasks, list_tasks, list_tasks, read_task_result, read_task_result, read_task_result, mongodb_aggregate, mongodb_find, mongodb_find, mongodb_aggregate, mongodb_find_one

Structured details:

```
{
  "stopReason": "max_tool_rounds",
  "toolCalls": [
    {
      "tool": "list_memory_keys"
    },
    {
      "tool": "mongodb_list_collections",
      "input": {
        "includeStats": true
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "emulator_spec_execution_state"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "emulator_execution_status"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "emulator_loop_status"
      }
    },
    {
      "tool": "mongodb_find_one",
      "input": {
        "collection": "convergence_state",
        "filter": "{}"
      }
    },
    {
      "tool": "mongodb_find_one",
      "input": {
        "collection": "sentinel_state",
        "filter": "{}",
        "sort": "{\"timestamp\": -1}"
      }
    },
    {
      "tool": "mongodb_find_one",
      "input": {
        "collection": "deployer_state",
        "filter": "{}"
      }
    },
    {
      "tool": "list_tasks",
      "input": {
        "status": "in_progress"
      }
    },
    {
      "tool": "list_tasks",
```

```
"input": {
  "status": "pending"
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "convergence_criteria",
    "filter": "{\"name\": \"insignia_subnet_convergence_v1\"}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "experiment_results",
    "filter": "{}",
    "sort": "{\"experiment_id\": -1}",
    "limit": 5,
    "detail_level": "L1"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "experiment_results",
    "detail_level": "L1",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 5,
    "sort": "{\"experiment_id\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "tuner_optimization",
    "detail_level": "L1",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\", \"version\": \"13.0-R3\"}",
    "limit": 3,
    "sort": "{\"generation\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "tuner_optimization",
    "detail_level": "L1",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 3,
    "sort": "{\"generation\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "experiment_results",
    "detail_level": "L0",
    "filter": "{}",
    "limit": 5,
    "sort": "{\"_id\": -1}"
  }
}
```

```

},
{
  "tool": "write_memory",
  "input": {
    "key": "emulator_spec_execution_state",
    "value": "{\"session\": \"EMULATOR_SPEC_execution_2026_06_30\", \"phase\": \"online_validation_dispatched\"}"
  }
},
{
  "tool": "write_task",
  "input": {
    "assignedTo": "deployer",
    "description": "The emulator loop is at Step 7 (empirical validation hold). Offline optimization has conver",
    "input": {
      "chain_endpoint": "ws://10.0.0.193:9945",
      "fallback_endpoint": "ws://10.0.0.193:9946",
      "knee_point_id": "V13-R3-KP-020-a3c7",
      "wallet_convention": {
        "owner": "insignia-owner",
        "validator": "insignia-validator",
        "miners": "insignia-miner-{0-11}"
      },
      "v13_r3_metrics": {
        "honest_score": 0.9808,
        "breach_rate": 0.0000026,
        "separation": 0.963,
        "variance": 0.00081
      }
    },
    "priority": 9,
    "title": "Deployer: Verify chain connectivity, fund wallets, deploy monitoring for online validation"
  }
},
{
  "tool": "write_task",
  "input": {
    "assignedTo": "coder",
    "description": "The sentinel reports 18/19 attack vectors below threshold. The single remaining warning is",
    "input": {
      "defense_id": "PC-VH-006",
      "target_vector": "sybil_attack",
      "target_repo": "sebastianrfelipe/Insignia-Subnet",
      "integration_points": [
        "tuning/attack_detector.py",
        "tuning/composite_integrity_scorer.py"
      ],
      "mongo_collection": "defense_strategies"
    },
    "priority": 8,
    "title": "Deploy PC-VH-006 Sybil defense (Symbol Diversity Enforcement) to clear last sentinel warning vect"
  }
},
{
  "tool": "mongodb_find_one",
  "input": {
    "collection": "defense_strategies",
    "filter": "{\"defense_id\": \"PC-VH-006\"}"
  }
},
{

```

```
"tool": "mongodb_find",
"input": {
  "collection": "breach_trends",
  "detail_level": "L1",
  "filter": "{}",
  "limit": 5,
  "sort": "{\"_id\": -1}"
}
},
{
  "tool": "get_details",
  "input": {
    "detail_level": "L2",
    "id": "6a419b17d163a13dc7ec918a",
    "resource_type": "mongodb_document"
  }
},
{
  "tool": "list_tasks",
  "input": {
    "status": "in_progress"
  }
},
{
  "tool": "read_task",
  "input": {
    "taskId": "6a419b17d163a13dc7ec918a"
  }
},
{
  "tool": "read_task",
  "input": {
    "taskId": "6a419b17d163a13dc7ec918c"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "sentinel_state",
    "detail_level": "L1",
    "filter": "{}",
    "limit": 3,
    "sort": "{\"_id\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "simulation_results",
    "detail_level": "L1",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 3,
    "sort": "{\"_id\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "composite_integrity_scores",
```

...(truncated)

[deployer] Deployer: Verify chain connectivity, fund wallets, deploy monitoring for online validation

Now let me update the deployer state in shared memory with the final comprehensive state:

Structured details:

```json

```
{
 "toolCallCount": 49,
 "toolCalls": [
 {
 "tool": "read_memory",
 "input": {
 "key": "deployer_state"
 }
 },
 {
 "tool": "read_memory",
 "input": {
 "key": "tuner_state"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "cat state/deployer_state.json 2>/dev/null || echo \"No local state file found\"",
 "description": "Read local deployer state"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "ls -la testnet/scripts/ 2>/dev/null || echo \"testnet/scripts/ not found\"; ls -la testnet/ 2>/dev/null || echo \"testnet/ not found\"",
 "description": "Check testnet directory structure"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "find /app -name \"check_chain_connectivity.sh\" -o -name \"check_wallet_balances.sh\" -o -name \"check_node_status.sh\" 2>/dev/null || echo \"No scripts found\"",
 "description": "Find testnet scripts anywhere in the project"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "docker ps -a 2>/dev/null | head -30; echo \"---\"; docker compose ps 2>/dev/null || docker-comp 2>/dev/null || echo \"No containers running\"",
 "description": "Check running Docker containers"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "ls -la /app/ | head -30",
 "description": "List app directory"
 }
 },
 {
 "tool": "shell_execute",

```

```
"input": {
 "command": "find /app -maxdepth 2 -type d 2>/dev/null | head -40",
 "description": "Find directory structure"
},
{
 "tool": "shell_execute",
 "input": {
 "command": "which btccli 2>/dev/null; btccli --version 2>/dev/null; which python3 2>/dev/null; python3 --vers
 "description": "Check bittensor tools availability"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "curl -s --connect-timeout 5 http://10.0.0.193:9945 2>&1 | head -5; echo \"EXIT: $?\"",
 "description": "Test connectivity to chain on port 9945"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "curl -s --connect-timeout 5 http://10.0.0.193:9946 2>&1 | head -5; echo \"EXIT: $?\"",
 "description": "Test connectivity to chain on port 9946"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "find /app/infrastructure -type f | head -40",
 "description": "List all files in infrastructure directory"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "which python3 || which python; which node; which npm; which npx; which wget; apt list --install
 "description": "Check available tools"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "ls -la /app/state/",
 "description": "Check state directory"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cat /app/infrastructure/chain/connectivity_check.mjs",
 "description": "Read chain connectivity check script"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cat /app/infrastructure/chain/check.mjs",
 "description": "Read chain check script"
 }
}
```

```

},
{
 "tool": "shell_execute",
 "input": {
 "command": "cat /app/infrastructure/wallets/verify.mjs",
 "description": "Read wallet verify script"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cat /app/infrastructure/monitoring/prometheus_server.mjs",
 "description": "Read monitoring script"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cat /app/state/v13r3_validation_results.json",
 "description": "Read V13-R3 validation results"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && node -e \"try { require('ws'); console.log('ws: available'); } catch(e) { console.lo",
 "description": "Check if ws package is available"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && node -e \"try { require('prom-client'); console.log('prom-client: available'); } cat",
 "description": "Check if prom-client is available"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"ws://10.0.0.193:9945\" node infrastructure/chain/connectivity_check",
 "description": "Test chain connectivity on port 9945",
 "timeout_ms": 15000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"ws://10.0.0.193:9946\" node infrastructure/chain/connectivity_check",
 "description": "Test chain connectivity on port 9946",
 "timeout_ms": 15000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"ws://127.0.0.1:9944\" node infrastructure/chain/connectivity_check.",
 "description": "Test chain connectivity on localhost 9944 (common default)",
 "timeout_ms": 15000
 }
},
},

```

```

{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"ws://127.0.0.1:9945\" node infrastructure/chain/connectivity_check.",
 "description": "Test chain connectivity on localhost 9945",
 "timeout_ms": 15000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"ws://127.0.0.1:9946\" node infrastructure/chain/connectivity_check.",
 "description": "Test chain connectivity on localhost 9946",
 "timeout_ms": 15000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"wss://entrypoint-finney.opentensor.ai:443\" node infrastructure/cha",
 "description": "Test connectivity to Finney mainnet",
 "timeout_ms": 20000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "which docker 2>/dev/null; docker --version 2>/dev/null; which docker-compose 2>/dev/null; docke",
 "description": "Check Docker availability and network interfaces"
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "node -e \"\nconst WebSocket = require('ws');\nconst endpoints = [\n 'ws://10.0.0.193:9944', 'w",
 "description": "Scan multiple endpoints for subtensor chain",
 "timeout_ms": 30000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"wss://test.finney.opentensor.ai:443\" node infrastructure/chain/con",
 "description": "Test connectivity to Finney testnet",
 "timeout_ms": 20000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"wss://entrypoint-finney.opentensor.ai:443\" node infrastructure/cha",
 "description": "Run full chain check with Finney fallback",
 "timeout_ms": 20000
 }
},
{
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && CHAIN_ENDPOINT=\"wss://entrypoint-finney.opentensor.ai:443\" node infrastructure/wal",
 "description": "Run wallet balance check against Finney",

```

```

 "timeout_ms": 20000
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "cd /app && node -e \"\nconst http = require('http');\nfunction check(url) {\n return new Promi
 \"description\": \"Check if Prometheus/Grafana already running\",
 \"timeout_ms\": 15000
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "node -e \"\nconst http = require('http');\nhttp.get('http://127.0.0.1:9090/metrics', res => {\n
 \"description\": \"Check Prometheus metrics endpoint\",
 \"timeout_ms\": 10000
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "node -e \"\nconst http = require('http');\nhttp.get('http://127.0.0.1:9090/health', res => {\n
 \"description\": \"Check Prometheus health endpoint\",
 \"timeout_ms\": 10000
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "ps aux 2>/dev/null | grep -E \"prometheus|grafana|node\" | grep -v grep | head -20",
 "description": "Check running processes"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "ls -la /app/infrastructure/config/ 2>/dev/null",
 "description": "Check infrastructure config directory"
 }
 },
 {
 "tool": "shell_execute",
 "input": {
 "command": "cat > /app/state/deployer_state.json << 'DEPLOYER_EOF'\n{\n \"version\": \"2.0\", \n \"procedu
... (truncated)

```

#### [tuner] CRITICAL: Investigate and fix score separation discrepancy (surrogate 0.963 vs empirical 0.244)

```

<tool_call>shell_execute<arg_key>command</arg_key><arg_value>cat << 'PYEOF' > /app/tuner_v13r3_diagnostic.py
#!/usr/bin/env python3
"""

```

TUNER V13-R3 DIAGNOSTIC: Score Separation Surrogate Failure Analysis

=====

Root cause analysis of why the surrogate GP model ( $R^2=0.96$ ) overestimated score\_separation by 4x (0.963 surrogate vs 0.244 empirical).

"""

```

import json
import numpy as np

```

```

from datetime import datetime

=====
1. EMPIRICAL DATA FROM MONGODB VALIDATION RUNS
=====

empirical_r1 = {
 "sim_id": "sim_v13r3_val_r1_1782533629474",
 "seed": 1042,
 "honest_mean_score": 0.9774,
 "best_adversarial_score": 0.7333,
 "score_separation": 0.2441,
 "chain_weights": {
 "Honest": 0.177, "HonestTrader": 0.174,
 "Copycat": 0.132, "CopyTrader": 0.131,
 "Overfitter": 0.108, "SingleMetricGamer": 0.108,
 "Sybil": 0.069, "Random": 0.099
 }
}

empirical_r2 = {
 "sim_id": "sim_v13r3_val_r2_1782533629474",
 "seed": 2042,
 "honest_mean_score": 0.9782,
 "best_adversarial_score": 0.7556,
 "score_separation": 0.2226,
 "chain_weights": {
 "Honest": 0.178, "HonestTrader": 0.178,
 "Copycat": 0.138, "CopyTrader": 0.131,
 "Overfitter": 0.108, "SingleMetricGamer": 0.105,
 "Sybil": 0.069, "Random": 0.094
 }
}

Surrogate predictions from the optimizer
surrogate_prediction = {
 "config_id": "V13-R3-KP-020-a3c7",
 "honest_score": 0.9808,
 "best_adversarial_score_surrogate_implied": None, # Will compute
 "score_separation": 0.963,
 "breach_rate": 0.0000026,
 "variance": 0.00081,
 "surrogate_r2": 0.96
}

=====
2. ROOT CAUSE ANALYSIS
=====

print("=" * 80)
print("TUNER V13-R3 DIAGNOSTIC: Score Separation Surrogate Failure Analysis")
print("=" * 80)

--- 2a. Compute what the surrogate IMPLIED for best_adversarial_score ---
If separation = honest - best_adversary, then:
best_adversary = honest - separation
surrogate_implied_best_adv = surrogate_prediction["honest_score"] - surrogate_prediction["score_separation"]
print(f"\n--- 2a. Surrogate-Implied vs Empirical Adversarial Scores ---")
print(f"Surrogate separation: {surrogate_prediction['score_separation']:.4f}")
print(f"Surrogate honest_score: {surrogate_prediction['honest_score']:.4f}")
print(f"Surrogate-implied best_adversary = {surrogate_prediction['honest_score']:.4f} - {surrogate_prediction['score_separation']:.4f}")

```

```

print(f"")
print(f"Empirical R1 best_adversary: {empirical_r1['best_adversarial_score']:.4f}")
print(f"Empirical R2 best_adversary: {empirical_r2['best_adversarial_score']:.4f}")
print(f"Empirical mean best_adversary: {np.mean([empirical_r1['best_adversarial_score'], empirical_r2['best_adversarial_score']):.4f}")
print(f"")
print(f"OVERESTIMATION FACTOR: surrogate separation / empirical separation = {surrogate_prediction['score_separation']:.4f}")
print(f"ADVERSARIAL SCORE UNDERESTIMATION: surrogate-implied ({surrogate_implied_best_adv:.4f}) vs actual ({np.mean([empirical_r1['best_adversarial_score'], empirical_r2['best_adversarial_score']):.4f})")

--- 2b. Separation metric definition analysis ---
print(f"\n--- 2b. Separation Metric Definition Analysis ---")
print(f"")
print(f"HYPOTHESIS 1: Optimizer uses separation = honest_score - best_adversarial_score")
sep_h1_r1 = empirical_r1["honest_mean_score"] - empirical_r1["best_adversarial_score"]
sep_h1_r2 = empirical_r2["honest_mean_score"] - empirical_r2["best_adversarial_score"]
print(f" R1: {empirical_r1['honest_mean_score']:.4f} - {empirical_r1['best_adversarial_score']:.4f} = {sep_h1_r1:.4f}")
print(f" R2: {empirical_r2['honest_mean_score']:.4f} - {empirical_r2['best_adversarial_score']:.4f} = {sep_h1_r2:.4f}")
print(f" MATCH: H1 matches empirical validator ✓")
print(f"")
print(f"HYPOTHESIS 2: Optimizer uses separation = honest_score / best_adversarial_score (ratio)")
sep_h2_r1 = empirical_r1["honest_mean_score"] / empirical_r1["best_adversarial_score"]
sep_h2_r2 = empirical_r2["honest_mean_score"] / empirical_r2["best_adversarial_score"]
print(f" R1: {empirical_r1['honest_mean_score']:.4f} / {empirical_r1['best_adversarial_score']:.4f} = {sep_h2_r1:.4f}")
print(f" R2: {empirical_r2['honest_mean_score']:.4f} / {empirical_r2['best_adversarial_score']:.4f} = {sep_h2_r2:.4f}")
print(f" MATCH: H2 does NOT match empirical (too high) X")
print(f"")
print(f"HYPOTHESIS 3: Surrogate modeled separation on log-transformed or normalized scale")
If the surrogate was trained on data where separation was ~0.95+ consistently,
it may have learned a mapping that doesn't extrapolate to the 0.2 range
print(f" Surrogate best_separation across Pareto front: 0.967 (max), 0.958-0.967 range")
print(f" Empirical separation: 0.22-0.24 range")
print(f" This suggests the surrogate NEVER saw a configuration with true separation < 0.9")
print(f" The GP surrogate is EXTRAPOLATING, not interpolating, at the true separation level")
print(f"")
print(f" CONCLUSION: The metric definition is CONSISTENT (honest - best_adv), but the surrogate model was trained on OPTIMIZER-EVALUATED separation values that were themselves inflated by a different evaluation procedure.")

--- 2c. WHY the surrogate overestimated ---
print(f"\n--- 2c. Root Cause: Why the Surrogate Overestimated by 4x ---")
print(f"")
print(f" ROOT CAUSE 1 (PRIMARY): Training data contamination")
print(f" The surrogate GP was trained on 630 evaluations from the NSGA-II optimizer.")
print(f" These evaluations used the OPTIMIZER's internal score_separation calculation,")
print(f" which differs from the empirical validator's calculation in key ways:")
print(f"")
print(f" a) The optimizer evaluates separation using a SIMPLIFIED adversary model:")
print(f" - It likely uses agent type scores from a fast analytical model")
print(f" - The analytical model may not capture the full adversarial strategy space")
print(f" - Copycat and CopyTrader scores in the analytical model: ~0.02-0.05")
print(f" - Copycat and CopyTrader scores in empirical validation: 0.132-0.138")
print(f" - This is a 3-7x underestimation of adversary capability")
print(f"")
print(f" b) The optimizer's honest_score calculation may use different aggregation:")
print(f" - Surrogate honest: 0.9808, Empirical honest: 0.9774-0.9782")
print(f" - Small difference (0.3%), so honest_score is approximately correct")
print(f" - The error is almost entirely in the ADVERSARIAL score estimation")
print(f"")
print(f" ROOT CAUSE 2 (SECONDARY): Surrogate extrapolation failure")
print(f" The GP surrogate with R²=0.96 fits the TRAINING data well, but the training data itself is biased. The R² metric measures fit to biased training targets,")

```

```

print(f" not accuracy against ground truth. This is a classic 'garbage in, garbage out'")
print(f" scenario where the surrogate faithfully reproduces a biased objective function.")
print(f"")
print(f" ROOT CAUSE 3 (CONTRIBUTING): Pareto front diversity collapse")
print(f" The Pareto front separation range is 0.958-0.967 (narrow 0.9% band).")
print(f" This means ALL 26 Pareto-optimal solutions have inflated separation,")
print(f" suggesting the optimizer converged to a false local optimum in the")
print(f" separation dimension.")

--- 2d. Chain weight analysis ---
print(f"\n--- 2d. Chain Weight Distribution Analysis ---")
for run_name, data in [("R1", empirical_r1), ("R2", empirical_r2)]:
 weights = data["chain_weights"]
 honest_total = weights["Honest"] + weights["HonestTrader"]
 adv_total = sum(v for k, v in weights.items() if k not in ["Honest", "HonestTrader"])
 print(f"\n {run_name} (seed {data['seed']}):")
 print(f" Honest weight total: {honest_total:.3f} ({honest_total*100:.1f}%)")
 print(f" Adversary weight total: {adv_total:.3f} ({adv_total*100:.1f}%)")
 print(f" Weight ratio (adv/honest): {adv_total/honest_total:.2f}x")
 print(f" Top adversary: Copycat={weights['Copycat']:.3f}, CopyTrader={weights['CopyTrader']:.3f}")
 print(f" Weakest adversary: Sybil={weights['Sybil']:.3f}")
 print(f"")
 print(f" CRITICAL INSIGHT: Adversaries hold {adv_total*100:.1f}% of weight!")
 print(f" This means the scoring system is NOT effectively separating honest from")
 print(f" adversarial miners. Copycat/CopyTrader are getting 26-27% of honest weight.")

--- 2e. Quantitative error decomposition ---
print(f"\n--- 2e. Quantitative Error Decomposition ---")
emp_mean_sep = np.mean([empirical_r1["score_separation"], empirical_r2["score_separation"]])
emp_mean_honest = np.mean([empirical_r1["honest_mean_score"], empirical_r2["honest_mean_score"]])
emp_mean_adv = np.mean([empirical_r1["best_adversarial_score"], empirical_r2["best_adversarial_score"]])

honest_error = surrogate_prediction["honest_score"] - emp_mean_honest
adv_error = surrogate_implied_best_adv - emp_mean_adv
sep_error = surrogate_prediction["score_separation"] - emp_mean_sep

print(f" Surrogate honest_score error: {honest_error:+.4f} ({honest_error/emp_mean_honest*100:+.2f}%)")
print(f" Surrogate adversarial score error: {adv_error:+.4f} ({adv_error/emp_mean_adv*100:+.2f}%)")
print(f" Surrogate separation error: {sep_error:+.4f} ({sep_error/emp_mean_sep*100:+.2f}%)")
print(f"")
print(f" Error attribution:")
print(f" Honest score error contribution: {abs(honest_error)/abs(sep_error)*100:.1f}% of separation error")
print(f" Adversarial score error contribution: {abs(adv_error)/abs(sep_error)*100:.1f}% of separation error")
print(f"")
print(f" CONCLUSION: {abs(adv_error)/abs(sep_error)*100:.1f}% of the separation overestimation")
print(f" comes from underestimating adversarial scores. The surrogate thought adversaries")
print(f" score ~0.018 when they actually score ~0.744 – a 41x underestimate.")

=====
3. PARAMETER SENSITIVITY ANALYSIS
=====
print(f"\n{'=' * 80}")
print(f"3. PARAMETER SENSITIVITY: What to Change to Increase Separation")
print(f"{'=' * 80}")

Current config
current_config = {
 "l1_weights": [0.24, 0.19, 0.14, 0.15, 0.12, 0.09, 0.07],
 "l2_weights": [0.2, 0.18, 0.14, 0.15, 0.13, 0.11, 0.09],
 "overfitting": {"gap_threshold": 0.16, "decay_rate": 0.94},

```

```

"promotion": {"top_n": 9, "min_consecutive_epochs": 4, "max_overfitting_score": 0.42,
 "max_score_decay_pct": 0.15, "expiry_epochs": 28},
"feedback": {"learning_rate": 0.012, "momentum": 0.58, "decay": 0.993, "clip": 1.15},
"anti_gaming": {"gaming_detection_sensitivity": 0.78, "penalty_escalation_rate": 0.65,
 "whitelist_decay_rate": 0.993, "cross_metric_correlation_threshold": 0.62},
"trading": {"min_trade_interval": 5, "max_position_concentration": 0.16,
 "slippage_tolerance": 0.01, "execution_timeout": 1100, "fill_rate_minimum": 0.9},
"emissions": {"base_emission_rate": 1.7, "performance_multiplier": 1.5,
 "longevity_bonus_rate": 0.14, "diversity_incentive_weight": 0.09,
 "decay_rate": 0.995, "min_emission_floor": 0.07, "max_emission_cap": 2.4},
"r3_features": {"decentralized_identity_verification": True, "bayesian_model_averaging": True,

```

...(truncated)

#### [researcher] Research: Diagnose why adversarial miners score 0.7333 (separation gate failure root cause)

Now I have the full picture. Let me also look at the middle part of the sim to understand chain weights computation

Structured details:

```

```json
{
  "toolCallCount": 26,
  "toolCalls": [
    {
      "tool": "read_memory",
      "input": {
        "key": "tuner_state"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "sentinel_state"
      }
    },
    {
      "tool": "read_memory",
      "input": {
        "key": "simulator_state"
      }
    },
    {
      "tool": "mongodb_list_collections",
      "input": {
        "includeStats": true
      }
    },
    {
      "tool": "mongodb_describe_collection",
      "input": {
        "collection": "simulation_agents",
        "sampleSize": 5
      }
    },
    {
      "tool": "mongodb_describe_collection",
      "input": {
        "collection": "simulation_epochs",
        "sampleSize": 3
      }
    }
  ]
}

```

```

},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "experiment_results",
    "sampleSize": 3
  }
},
{
  "tool": "mongodb_describe_collection",
  "input": {
    "collection": "chain_weights",
    "sampleSize": 5
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "simulation_results",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 10,
    "sort": "{\"timestamp\": -1}"
  }
},
{
  "tool": "mongodb_find",
  "input": {
    "collection": "chain_weights",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\"}",
    "limit": 10,
    "sort": "{\"timestamp\": -1}"
  }
},
{
  "tool": "mongodb_aggregate",
  "input": {
    "collection": "experiment_results",
    "pipeline": "[{\"$match\": {\"procedure\": \"insignia_subnet_tuner\", \"status\": \"completed\"}}, {\"$sort
  }
},
{
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  "input": {
    "collection": "simulation_epochs",
    "filter": "{\"simulation_id\": {\"$in\": [\"sim_v13r3_val_r1_1782533629474\", \"sim_v13r3_val_r2_1782533629
    \"limit\": 20,
    \"sort\": \"{\"epoch\": -1}\"
  }
},
{
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  "input": {
    "collection": "experiment_results",
    "filter": "{\"procedure\": \"insignia_subnet_tuner\", \"target\": {\"$regex\": \"separation|copycat|copy|co
    \"limit\": 20,
    \"sort\": \"{\"timestamp\": -1}\"
  }
},
{
  "tool": "mongodb_aggregate",

```

```

    "input": {
      "collection": "experiment_results",
      "pipeline": "[{\"$match\": {\"procedure\": \"insignia_subnet_tuner\"}}, {\"$sort\": {\"timestamp\": -1}}, {
    },
    {
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      "input": {
        "collection": "experiment_results",
        "pipeline": "[{\"$sort\": {\"timestamp\": -1}}, {\"$limit\": 15}, {\"$project\": {\"experiment_id\": 1, \"c
      }
    },
    {
      "tool": "mongodb_find",
      "input": {
        "collection": "simulation_agents",
        "filter": "{ \"simulation_id\": { \"$in\": [ \"sim_v13r3_val_r1_1782533629474\", \"sim_v13r3_val_r2_1782533629
        \"limit\": 20
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find / -path \"*/tuning/*\" -name \"*.py\" 2>/dev/null | head -30",
        "timeout_ms": 15000
      }
    },
    {
      "tool": "mongodb_aggregate",
      "input": {
        "collection": "simulation_agents",
        "pipeline": "[{\"$match\": {\"simulation_id\": \"sim_v13r3_val_r1_1782533629474\", \"epoch\": 1, \"adversar
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find /app -name \"*.py\" -path \"*/tuning/*\" 2>/dev/null; find /app -name \"autoresearch_loop.
        "timeout_ms": 15000
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "find /app -name \"*.py\" 2>/dev/null | head -50",
        "timeout_ms": 15000
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "ls -la /app/tuner/ && ls -la /app/src/ && ls -la /app/results/",
        "timeout_ms": 10000
      }
    },
    {
      "tool": "shell_execute",
      "input": {
        "command": "cat /app/sim_v13r3.mjs | head -200",
        "timeout_ms": 10000

```

```

    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "cat /app/sim_v13r3.mjs | tail -100",
      "timeout_ms": 10000
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "cat /app/sim_v13r3.mjs | wc -l",
      "timeout_ms": 5000
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "sed -n '200,360p' /app/sim_v13r3.mjs",
      "timeout_ms": 10000
    }
  },
  {
    "tool": "shell_execute",
    "input": {
      "command": "cd /app && git status 2>&1; git remote -v 2>&1; git log --oneline -5 2>&1",
      "timeout_ms": 10000
    }
  }
],
"usage": {
  "inputTokens": 189495,
  "outputTokens": 10232
},
"stopReason": "length",
"throughput": {
  "llmDurationMs": 30531,
  "outputTokensPerSec": 335.1,
  "totalTokensPerSec": 6541.7,
  "rounds": [
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      "round": 1,
      "durationMs": 1046,
      "inputTokens": 3324,
      "outputTokens": 152,
      "outputTokensPerSec": 145.3
    },
    {
      "round": 2,
      "durationMs": 1846,
      "inputTokens": 5787,
      "outputTokens": 264,
      "outputTokensPerSec": 143
    },
    {
      "round": 3,
      "durationMs": 3208,
      "inputTokens": 13546,
      "outputTokens": 522,

```

```
    "outputTokensPerSec": 162.7
  },
  {
    "round": 4,
    "durationMs": 6007,
    "inputTokens": 16644,
    "outputTokens": 311,
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  },
  {
    "round": 5,
    "durationMs": 4167,
    "inputTokens": 17982,
    "outputTokens": 226,
    "outputTokensPerSec": 54.2
  },
  {
    "round": 6,
    "durationMs": 3060,
    "inputTokens": 21772,
    "outputTokens": 284,
    "outputTokensPerSec": 92.8
  },
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    "round": 7,
    "durationMs": 2908,
    "inputTokens": 23174,
    "outputTokens": 75,
    "outputTokensPerSec": 25.8
  },
  {
    "round": 8,
    "durationMs": 2286,
    "inputTokens": 27439,
    "outputTokens": 87,
    "outputTokensPerSec": 38
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  {
    "round": 9,
    "durationMs": 3188,
    "inputTokens": 28732,
    "outputTokens": 119,
    "outputTokensPerSec": 37.3
  },
  {
    "round": 10,
    "durationMs": 2816,
    "inputTokens": 31095,
    "outputTokens": 8192,
    "outputTokensPerSec": 2909.5
  }
]
}
```

Errors (1)

[coder] Deploy PC-VH-006 Sybil defense (Symbol Diversity Enforcement) to clear last sentinel warning vector

Task was cancelled before completion.

Insights (2)

- **[deployer] Deployer State:** {
"version": "3.0",
"procedure": "insignia_subnet_tuner",
"phase": "online_transition_blocked",
"emulator_step": 7,
"knee_point": "V13-R3-KP-020-a3c7",
"timestamp": "2026-06-28T22:18:00Z",
"chain_connectivity": {
"SUBSENSOR_LOCAL_ENDPOINT": "ws://10.0.0.193:9945",
"port_9945_result": "ECONNREFUSED",
"port_9946_result": "ECONNREFUSED",
"legacy_9933_result": "ECONNREFUSED (misconfigured, per EMULATOR_SPEC §4 should be 9945)",
"all_local_endpoints_unreachable": true
}
}
- **[orchestrator] Emulator Spec Execution State:** {"session": "EMULATOR_SPEC_execution_2026_06_30", "phase": "separation_gap_investigation", "timestamp": "2026-07-17T14:15:00Z", "critical_finding": "V13-R3 FAILS score_separation acceptance gate. Surrogate reported 0.963 separation, but empirical validation shows 0.244/0.223. Best adversarial score is 0.7333 (target: near-zero). Adversaries capture 64.7% of chain weight. Production promotion BLOCKED.", "system_state": {"tuner": {"knee_point": "V13-R3-KP-020-a3c7", "convergence_trend": "INVALIDAT

Task Details

Status	Agent	Title	Result			
Done	orchestrator	Execute skill: EMULATOR_SPEC	Now I have the per-archetype score breakdown. This is very revealing:	Archetype	Adversarial	A
Done	deployer	Deployer: Verify chain connectivity, fund wallets, deploy monitoring for online validation	Now let me update the deployer state in shared memory with the final comprehensive state:			
cancelled	coder	Deploy PC-VH-006 Sybil defense (Symbol Diversity Enforcement) to clear last sentinel warning vector				
Done	tuner	CRITICAL: Investigate and fix score separation discrepancy (surrogate 0.963 vs empirical 0.244)	<tool_call>shell_execute<arg_key>command</arg_key><arg_value>cat << 'PYEOF' > /app/tuner_v13r3_diagn			
Done	researcher	Research: Diagnose why adversarial miners score 0.7333 (separation gate failure root cause)	Now I have the full picture. Let me also look at the middle part of the sim to understand chain weig			

Tool Usage

Tool	Calls
shell_execute	54
mongodb_find	33
read_memory	12
mongodb_find_one	10
list_tasks	6
mongodb_aggregate	6
mongodb_list_collections	4
write_memory	4
write_task	4
mongodb_describe_collection	4
read_task_result	3
get_details	2
read_task	2
dashboard_skill_execute	1
list_memory_keys	1
cancel_task	1

Errors: 2

Agent Memory

Deployer State

Written by deployer

```

{
  "version": "3.0",
  "procedure": "insignia_subnet_tuner",
  "phase": "online_transition_blocked",
  "emulator_step": 7,
  "knee_point": "V13-R3-KP-020-a3c7",
  "timestamp": "2026-06-28T22:18:00Z",
  "chain_connectivity": {
    "SUBTENSOR_LOCAL_ENDPOINT": "ws://10.0.0.193:9945",
    "port_9945_result": "ECONNREFUSED",
    "port_9946_result": "ECONNREFUSED",
    "legacy_9933_result": "ECONNREFUSED (misconfigured, per EMULATOR_SPEC §4 should be 9945)",
    "all_local_endpoints_unreachable": true,
    "docker_available_in_container": false,
    "container_ip": "172.17.0.3",
    "finney_mainnet": {
      "endpoint": "wss://entrypoint-finney.opentensor.ai:443",
      "status": "CONNECTED",
      "block_height": 8507775,
      "chain_name": "Bittensor",
      "is_syncing": false,
      "peers": 40
    },
    "finney_testnet": {
      "endpoint": "wss://test.finney.opentensor.ai:443",
      "status": "CONNECTED",
      "block_height": 7446145,
      "chain_name": "Bittensor",
      "is_syncing": false,
      "peers": 24
    },
    "diagnosis": "Pre-provisioned Docker subtensor at 10.0.0.193 is NOT running or network-isolated from this conta
  },
  "wallet_status": {
    "total_wallets": 14,
    "all_funded": false,
    "total_funding_needed_tao": 112000,
    "breakdown": {
      "insignia-owner": {
        "role": "subnet_owner",
        "needed": 50000
      },
      "insignia-validator": {
        "role": "validator",
        "needed": 50000
      },
      "insignia-miner-0_through_11": {
        "role": "miner",
        "needed_each": 1000,
        "count": 12,
        "subtotal": 12000
      }
    },
    "note": "All wallets show zero balance. Cannot fund without local subtensor faucet. Funding blocked until local
  },
  "subnet_registration": {
    "status": "BLOCKED",

```

"rea
...(truncated)

Emulator Spec Execution State

Written by orchestrator

{"session": "EMULATOR_SPEC_execution_2026_06_30", "phase": "separation_gap_investigation", "timestamp": "2026-07-17"
...(truncated)



Report generated by Insignia Swarm Gateway